

Marketing analytics overview



 **BA UNIT 4.pdf**
PDF

Explain marketing analytics and its importance in business? L2 CO4 10M

2 A Discuss the different market analytics models? L2 CO4 10M

3 A Discuss in detail the measurement of marketing metrics. L2 CO4 10M

4 A What are the different sources for market data? Explain in brief. L1 CO4 10M

5

A Explain market sizing techniques. L2 CO4 5M

B Discuss sales metrics used in sales analytics? L3 CO4 5M

6 A Discuss in detail about PESTEL analysis L3 CO4 10M

7

A Explain porter 5 forces in marketing analytics. L2 CO4 5M

B What is text analytics? Explain the benefits. L1 CO4 5M

8

A Discuss an analysis technique for understanding customer buying behaviour
Explain market basket analytics ?

Here's a complete explanation for each of your questions from **BA Unit 4 (Market Analytics, Models & Metrics)** — simplified and clearly structured for your 10M & 5M answers  

1. Explain Marketing Analytics and its Importance in Business (L2 CO4 – 10M)

Definition:

Marketing analytics is the process of using **data and statistical analysis** to evaluate the effectiveness of marketing activities. It helps businesses make **data-driven decisions** to improve performance, optimize campaigns, and maximize ROI.

How it works:

1. Identify what you want to measure (e.g., ROI, conversion rate).
2. Use analytic tools to analyze past, present, and future performance.
3. Fill analytical gaps with suitable models and techniques.
4. Act on the insights to optimize strategy.

Importance in Business:

- Helps understand customer behavior and preferences.
- Optimizes marketing budgets and resources.

- Increases ROI by identifying what works.
 - Improves customer targeting and personalization.
 - Enhances long-term business planning and competitiveness.
 - Enables better supply-demand planning and lead nurturing.
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2. Discuss the Different Market Analytics Models (L2 CO4 – 10M)

1. Customer Acquisition Modelling:

Identifies which prospects are most likely to become customers.

→ Example: A non-profit predicting which visitors will donate.

2. Recommendation Modelling:

Suggests relevant or higher-value products.

- **Upsell:** Predicts customers likely to buy premium products.
- **Cross-sell:** Finds customers likely to buy from other categories.
- **Next Best Offer:** Suggests personalized product recommendations.

3. Fast-track Modelling:

Predicts which new customers will become high-value clients over time.

4. Churn Modelling:

Identifies customers likely to stop using a product/service.

5. Slider Modelling:

Detects customers who may shift to competitors.

6. Customer Commitment Modelling:

Measures customer loyalty and satisfaction to target specific groups better.

3. Discuss in Detail the Measurement of Marketing Metrics (L2 CO4 – 10M)

Definition:

Marketing metrics are **quantifiable measures** used to track marketing performance and ROI.

Important Marketing Metrics:

1. **Traffic by Source** – Identifies where visitors come from (SEO, ads, etc.)
2. **Unique Monthly Visitors** – Tracks reach and audience size.
3. **Cost per Lead (CPL)** – Marketing spend / Number of leads.
4. **Cost per Acquisition (CPA)** – Cost to gain a paying customer.
5. **Conversion Rate** – % of users who complete a desired action.
6. **Return on Advertising Spend (ROAS)** – Revenue / Ad spend.
7. **Page Performance** – Measures engagement and speed.
8. **Customer Lifetime Value (CLV)** – Total revenue per customer over time.
9. **Average SERP Position** – Measures SEO ranking.

10. **Branded Search Queries** – Tracks brand awareness over time.

These metrics guide marketers in optimizing campaigns and achieving business goals.

4. What are the Different Sources for Market Data? (L1 CO4 – 10M)

Market Data Sources Include:

1. Primary Data Sources:

- Surveys, interviews, focus groups.
- Direct observation and experiments.

2. Secondary Data Sources:

- Government publications (Census, RBI, etc.).
- Industry reports, company websites, and market research firms.
- Social media analytics, CRM systems, and sales data.

3. Digital Data Sources:

- Web analytics tools (Google Analytics, SEMrush).
- E-commerce data, ad campaign performance, search trends.

Purpose:

These sources provide insights into consumer behavior, competition, and market size — essential for making data-driven business strategies.

5A. Explain Market Sizing Techniques (L2 CO4 – 5M)

Definition:

Market sizing estimates the potential reach and revenue for a product/service.

Key Types:

1. **Market Potential:** 100% market share — total possible value.
2. **TAM (Total Addressable Market):** Entire demand for the product.
3. **SAM (Serviceable Available Market):** The portion your business can serve.
4. **SOM (Serviceable Obtainable Market):** The share your company can realistically capture.

Approaches:

- **Top-down:** Starts from macro data (population, industry data).
- **Bottom-up:** Starts from individual or micro-level estimates and scales up.

Importance:

Helps in identifying opportunities, validating business models, and attracting investors.

5B. Discuss Sales Metrics Used in Sales Analytics (L3 CO4 – 5M)

Common Sales Metrics:

1. **Sales Growth:** Measures increase in revenue over time.
2. **Sales Target Achievement:** Tracks performance vs. goals.
3. **Sales per Rep:** Evaluates individual performance.
4. **Sales by Region:** Assesses sales in different territories.
5. **Sell-through Rate:** Measures how quickly inventory sells.
6. **Pipeline Velocity:** Tracks revenue flow in the sales funnel.
7. **Quote to Close:** % of leads converted to sales.
8. **Average Purchase Value:** Average revenue per transaction.

These metrics help managers improve team efficiency, forecasting, and profitability.

6. Discuss in Detail about PESTEL Analysis (L3 CO4 – 10M)

Definition:

PESTEL is a tool to analyze the **macro-environmental factors** influencing a business. It stands for **Political, Economic, Social, Technological, Legal, Environmental**.

Components:

- **Political:** Government policies, tax laws, trade tariffs.
- **Economic:** Inflation, interest rates, currency exchange, growth trends.
- **Social:** Culture, demographics, lifestyle, consumer attitudes.
- **Technological:** Innovation, automation, R&D advancements.
- **Legal:** Labor laws, safety standards, consumer protection.
- **Environmental:** Climate change, sustainability, pollution norms.

Importance:

Helps organizations identify external threats/opportunities and make informed strategic decisions.

7A. Explain Porter's Five Forces in Marketing Analytics (L2 CO4 – 5M)

Definition:

Michael Porter's model analyzes **industry competitiveness** to determine profitability.

Five Forces:

1. **Competition in the Industry:** More rivals = less market power.
2. **Threat of New Entrants:** Low entry barriers increase competition.
3. **Power of Suppliers:** Few suppliers = higher bargaining power.
4. **Power of Customers:** More buyers' options = lower prices.
5. **Threat of Substitutes:** Alternative products reduce demand.

Purpose:

Helps businesses analyze competition and develop strategies for long-term profitability.

7B. What is Text Analytics? Explain the Benefits (L1 CO4 – 5M)

Definition:

Text analytics uses **machine learning and linguistic techniques** to process and analyze large volumes of unstructured text data to extract meaningful insights.

Benefits:

- Understand customer opinions (via sentiment analysis).
- Identify trends from feedback or reviews.
- Improve customer experience with data-driven actions.
- Helps in risk monitoring, fraud detection, and product improvement.
- Reduces human bias in data interpretation.

Example:

Analyzing social media comments or customer reviews to understand brand perception.

8A. Discuss an Analysis Technique for Understanding Customer Buying Behaviour – Explain Market Basket Analytics

Market Basket Analysis (MBA):

A data mining technique that finds **associations between products frequently bought together**. It's based on **association rule mining** (IF {A} THEN {B}).

Key Terms:

- **Antecedent:** Item(s) first bought (e.g., Bread).
- **Consequent:** Item(s) bought with it (e.g., Butter).

Key Measures:

- **Support:** Frequency of item combination in transactions.
- **Confidence:** Likelihood of buying B after A.
- **Lift:** Strength of association between items.

Example:

"If customers buy bread, they are likely to buy butter."

Amazon uses this to suggest "Frequently Bought Together" items.

Benefits:

- Enhances cross-selling and up-selling.
- Improves product placement in stores.
- Boosts sales and customer satisfaction.

Would you like me to make this into a **formatted PDF (exam-ready notes)** version for printing and submission?



 **BA_UNIT 5 (1).pdf**
PDF

What is Big Data? Explain the characteristics of Big Data L2 CO5 5 M
B Discuss on master data management (MDM)? L3 CO5 5 M

2

A What is data mining and explain the steps involved in mining? L2 CO5 5 M
B What are the benefits of data mining? L2 CO5 5 M

3

A Explain the kinds of data mining tools available? Discuss the applications of data mining.

L3 CO5 10 M

4 A Explain data objects & Attributes? Explain types of Attributes L2 CO5 10 M

5

A List out the challenges/major issues in data mining. L2 CO5 5 M

B Explain the different kinds of patterns can be mined? L3 CO5 5 M

6 A Explain the technologies and applications used in data mining? L3 CO5 10 M

7 A Explain the basic statistical descriptions of data? L2 CO5 10 M

8 A Explain the data similarity and dissimilarity in detail.

Perfect 💡 I went through your uploaded **BA Unit 5 PDF**, and here's a summarized, exam-ready explanation for each of your questions from **CO5 (Big Data & Data Mining Unit)** — clear and concise for 5M and 10M answers:

1A. What is Big Data? Explain the characteristics of Big Data (L2, 5M)

Definition:

Big Data refers to extremely large and complex data sets that traditional data processing tools cannot handle efficiently. It involves collecting, storing, and analyzing massive amounts of structured, semi-structured, and unstructured data to derive insights.

Characteristics (5 V's of Big Data):

1. **Volume** – The vast amount of data generated every second from sensors, social media, IoT, etc.
2. **Velocity** – The speed at which data is generated, collected, and processed in real-time.
3. **Variety** – Data comes in multiple formats: text, audio, video, images, etc.
4. **Veracity** – Ensures the accuracy and reliability of the data collected.

5. **Value** – The most important V; the ability to derive meaningful insights that create business value.

1B. Discuss Master Data Management (MDM) (L3, 5M)

Definition:

Master Data Management (MDM) is a process and technology framework ensuring the uniformity, accuracy, and consistency of an organization's key data assets like customers, products, suppliers, and locations.

Key Points:

- MDM creates a **"single source of truth"** or "golden record" for enterprise data.
- Integrates structured and unstructured data across multiple systems.
- Ensures data consistency and improves business analytics accuracy.

Benefits:

- Enhances data quality and governance.
 - Improves customer experience and operational efficiency.
 - Enables better decision-making and compliance with data regulations.
-

2A. What is Data Mining? Explain the steps involved (L2, 5M)

Definition:

Data Mining is the process of discovering hidden patterns, correlations, and useful information from large data sets using statistical, machine learning, and AI techniques.

Steps in Data Mining:

1. **Business Understanding** – Define objectives and success criteria.
 2. **Data Understanding** – Identify and collect relevant data.
 3. **Data Preparation** – Clean and transform raw data.
 4. **Modeling** – Apply algorithms to identify patterns or build predictive models.
 5. **Evaluation** – Validate the model's accuracy and effectiveness.
 6. **Deployment** – Implement the results in decision-making systems.
-

2B. Benefits of Data Mining (L2, 5M)

- Detects fraud and reduces risks.
 - Improves decision-making using data-driven insights.
 - Identifies hidden patterns and customer behavior trends.
 - Enhances marketing strategies through segmentation.
 - Supports predictive maintenance and forecasting.
-

3A. Kinds of Data Mining Tools and Applications (L3, 10M)

Tools:

- **AI and Machine Learning Tools** – e.g., TensorFlow, Scikit-learn.
- **Data Analytics Tools** – RapidMiner, KNIME, Weka.
- **Database Tools** – Oracle Data Mining, SQL Server Analysis Services.
- **Statistical Tools** – R, SAS, SPSS.
- **Big Data Tools** – Hadoop, Spark, NoSQL databases.

Applications:

- **Banking** – Loan prediction, fraud detection.
 - **Healthcare** – Disease prediction, patient profiling.
 - **Retail** – Market basket analysis, customer segmentation.
 - **Telecom** – Detecting fraud patterns and churn prediction.
 - **Manufacturing** – Predictive maintenance, process optimization.
-

4A. Data Objects & Attributes (L2, 10M)

- **Data Objects:** Represent entities such as people, events, or things described by attributes.
- **Attributes:** The characteristics or properties of a data object.

Types of Attributes:

1. **Nominal** – Categories (e.g., gender: male/female).
 2. **Ordinal** – Ordered categories (e.g., rank: high, medium, low).
 3. **Interval** – Numerical with meaningful differences but no true zero (e.g., temperature in °C).
 4. **Ratio** – Numeric with a true zero (e.g., height, weight, salary).
-

5A. Challenges in Data Mining (L2, 5M)

- **Data Quality Issues** – Noisy, incomplete, inconsistent data.
 - **Scalability** – Handling massive datasets.
 - **Data Security and Privacy.**
 - **Complex Data Types** – Multimedia, time-series, spatial.
 - **Integration from Multiple Sources.**
-

5B. Different Kinds of Patterns that can be Mined (L3, 5M)

1. **Association Patterns** – Items that occur together (e.g., market basket analysis).
2. **Sequential Patterns** – Time-based sequences (e.g., customer purchase history).

3. **Classification Patterns** – Assign items to predefined categories.
 4. **Clustering Patterns** – Group similar data points.
 5. **Prediction Patterns** – Forecast future trends.
-

6A. Technologies & Applications Used in Data Mining (L3, 10M)

Technologies:

- Hadoop & Spark (Big Data Frameworks).
- NoSQL Databases (MongoDB, Cassandra).
- Data Warehouses.
- Machine Learning & AI Models.
- Cloud Platforms (AWS, Google Cloud, Azure).

Applications:

- Fraud Detection (Banking).
 - Disease Diagnosis (Healthcare).
 - Market Segmentation (Retail).
 - Network Optimization (Telecom).
 - Predictive Analytics (Manufacturing).
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7A. Basic Statistical Descriptions of Data (L2, 10M)

- **Mean, Median, Mode** – Central tendencies.
 - **Range, Variance, Standard Deviation** – Data dispersion.
 - **Skewness & Kurtosis** – Shape of data distribution.
 - **Correlation & Covariance** – Relationships between attributes.
-

8A. Data Similarity and Dissimilarity (L2, 10M)

- **Similarity:** Measures how alike two objects are.
 - Common Measures: Cosine similarity, Jaccard index, Pearson correlation.
 - **Dissimilarity:** Measures how different two objects are.
 - Common Measures: Euclidean distance, Manhattan distance, Hamming distance.
Used in clustering and pattern recognition to group or differentiate datasets.
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Would you like me to make this into a clean **exam notes PDF** (with 5M/10M formatting and neat headings)? It'll look good for quick revision before your test.



Explain the features of Big Data?

L2 CO5

2M

2 List the types of Big data analytics

L3 CO5

2M

3 List the components of Master Data?

L2 CO5

2M

4 Explain Master Data Governance

L2 CO5

2M

5 List data mining steps

L3 CO5

2M

6 List two challenges in Data mining?

L2 CO5

2M

7 Data Objects Vs. Attributes

L2 CO5

2M

8 Explain any two Data mining tools

L3 CO5

2M

9 Explain affinity analysis?

L2 CO5

2M

10

Data similarity Vs. Date dissimilarity

Here's your **2 Marks short answers (CO5 – Big Data & Data Mining Unit)** — concise and exam-friendly



1. Explain the features of Big Data (L2 CO5 – 2M)

- **High Volume:** Massive amounts of data generated from multiple sources.
 - **High Velocity:** Data flows in real time and must be processed quickly.
 - **Variety:** Comes in multiple formats—structured, semi-structured, and unstructured.
 - **Veracity:** Data quality and reliability matter.
 - **Value:** Extracting meaningful insights from data creates business value.
-

2. List the types of Big Data Analytics (L3 CO5 – 2M)

1. **Descriptive Analytics** – Describes what has happened.
 2. **Diagnostic Analytics** – Explains why it happened.
 3. **Predictive Analytics** – Forecasts what might happen.
 4. **Prescriptive Analytics** – Suggests actions to be taken.
-

3. List the components of Master Data (L2 CO5 – 2M)

- **Customer Data**
 - **Product Data**
 - **Supplier Data**
 - **Location Data**
 - **Employee Data**
-

4. Explain Master Data Governance (L2 CO5 – 2M)

It refers to the **policies, rules, and procedures** that ensure master data is **accurate, consistent, and compliant** across the organization.

Governance defines ownership, quality standards, and accountability for managing data assets.

5. List Data Mining Steps (L3 CO5 – 2M)

1. Business Understanding
2. Data Understanding
3. Data Preparation
4. Modeling
5. Evaluation
6. Deployment

6. List Two Challenges in Data Mining (L2 CO5 – 2M)

- Handling large and complex datasets (Scalability).
- Ensuring data quality, privacy, and security.

7. Data Objects vs Attributes (L2 CO5 – 2M)

Data Objects	Attributes
Represent entities like a person, product, or event.	Describe the properties or features of data objects.
Example: Student	Example: Name, Age, Grade

8. Explain Any Two Data Mining Tools (L3 CO5 – 2M)

1. **RapidMiner:** Open-source tool for predictive modeling and text mining.
2. **Weka:** Machine learning software used for data pre-processing and classification.

9. Explain Affinity Analysis (L2 CO5 – 2M)

Affinity analysis identifies **relationships between items or events** that occur together.

Example: "Customers who buy bread also buy butter."

(Used in Market Basket Analysis.)

10. Data Similarity vs Data Dissimilarity (L2 CO5 – 2M)

Similarity	Dissimilarity
Measures how alike two objects are.	Measures how different two objects are.
Higher value = more similar.	Higher value = more dissimilar.
Example: Cosine Similarity.	Example: Euclidean Distance.

Would you like me to make a **one-page PDF** version of these 2-mark answers for quick print/revision before exams?



What is Marketing Analytics?

L2 CO4

2M

2 Upselling Vs. Cross Selling

L3 CO4

2M

3

List two marketing metrics for establishing dashboard of marketing analytics?

L2 CO4

2M

4 Explain Marketing data resources

L3 CO4

2M

5 Accronym of PESTEL

L3 CO4

2M

6 List Porter's five forces

L3 CO4

2M

7 Explain the Objective of Market-Basket analysis

L2 CO4

2M

8 What is Market sizing?

L3 CO4

2M

9 List the types of Marketing metrics

L2 CO4

2M

10 Explain Text analytics

Here's your **2 Marks Short Answers (CO4 – Marketing Analytics Unit)** — quick, neat, and exam-ready ☐

1. What is Marketing Analytics? (L2 CO4 – 2M)

Marketing analytics is the process of **collecting, measuring, and analyzing marketing data** to evaluate performance, understand customer behavior, and improve marketing decisions using data-driven insights.

2. Upselling vs Cross-Selling (L3 CO4 – 2M)

Upselling	Cross-Selling
Encouraging a customer to buy a higher-end or premium product.	Suggesting related or complementary products.
Example: Recommending a costlier phone model.	Example: Suggesting a phone case with the phone.

3. List Two Marketing Metrics for Establishing Dashboard of Marketing Analytics (L2 CO4 – 2M)

- **Cost per Lead (CPL)**
- **Customer Lifetime Value (CLV)**

(Also possible: Conversion Rate, ROI, or Click-Through Rate)

4. Explain Marketing Data Resources (L3 CO4 – 2M)

Marketing data resources include **internal and external sources** such as:

- Internal: CRM systems, sales data, website analytics.
- External: Market research reports, social media, government data.

5. Acronym of PESTEL (L3 CO4 – 2M)

P – Political
E – Economic
S – Social
T – Technological
E – Environmental
L – Legal

6. List Porter's Five Forces (L3 CO4 – 2M)

1. Rivalry among existing competitors
2. Threat of new entrants
3. Bargaining power of suppliers

4. Bargaining power of customers
5. Threat of substitute products

7. Explain the Objective of Market Basket Analysis (L2 CO4 – 2M)

The main objective is to **identify products that are frequently bought together**, helping businesses improve **cross-selling and store layout** for higher sales.

8. What is Market Sizing? (L3 CO4 – 2M)

Market sizing estimates the **total potential sales or revenue opportunity** for a product or service within a specific market.

9. List the Types of Marketing Metrics (L2 CO4 – 2M)

1. **Input Metrics** – Measure marketing efforts (e.g., ad spend).
2. **Output Metrics** – Measure results (e.g., leads, conversions).
3. **Outcome Metrics** – Measure business impact (e.g., ROI, CLV).

10. Explain Text Analytics (L2 CO4 – 2M)

Text analytics is the process of **analyzing unstructured text data** (like reviews or social media posts) to extract insights such as **sentiments, trends, and key topics** using NLP and AI.

Would you like me to merge both **CO4 and CO5 2-mark answers** into a single printable **summary sheet (PDF)** for revision?



Essay type Questions

Level of
Bloom
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my
CO
Marks
(5/10
M)

- 1 A What is data visualization and Explain its importance in business analytics? L2 CO3 4M
- 2 A Discuss Visual Analytics L3 CO3 4M
- 3 A What is linear optimization model? Explain its process steps. L2 CO3 4M
- 4 A Explain features of linear programming? L3 CO3 4M
- 5 A Discuss Implementation of Linear Optimization models on spreadsheets? L3 CO3 4M
- 6 A Discuss the Solving procedures using Linear Optimization models

Here's a neat, exam-ready summary of your **CO3 – Essay Type Questions (Data Visualization & Linear Optimization)** 🌸

1A. What is Data Visualization and Explain its Importance in Business Analytics (L2 CO3 – 4M)

Definition:

Data visualization is the process of **representing data and information through graphical formats** such as charts, graphs, maps, and dashboards to make complex data more understandable.

Importance in Business Analytics:

- **Simplifies Complex Data:** Converts raw data into easy-to-read visuals.
- **Improves Decision-Making:** Helps managers spot trends and patterns quickly.
- **Enhances Communication:** Makes reports visually appealing and clear.
- **Identifies Trends & Outliers:** Detects anomalies that may affect performance.
- **Supports Real-Time Analysis:** Dashboards allow continuous business monitoring.

Example:

A company visualizes monthly sales using bar charts to identify top-performing regions.

2A. Discuss Visual Analytics (L3 CO3 – 4M)

Definition:

Visual analytics combines **data analytics and interactive visualizations** to help users explore data, detect patterns, and derive insights intuitively.

Key Features:

- Interactive dashboards and dynamic visual tools.
- Integrates **data mining, statistics, and visualization**.
- Supports **real-time decision-making** through visual exploration.

Advantages:

- Improves understanding of large datasets.
- Enables human-computer collaboration in analysis.
- Detects hidden relationships and patterns in data.

Example:

Analyzing customer segments through interactive dashboards in Tableau or Power BI.

3A. What is Linear Optimization Model? Explain Its Process Steps (L2 CO3 – 4M)

Definition:

A **linear optimization model (linear programming)** is a mathematical method used to **maximize or**

minimize a linear objective function subject to linear constraints.

Process Steps:

1. **Define the Decision Variables** – What needs to be determined (e.g., quantity to produce).
2. **Formulate the Objective Function** – Goal to maximize profit or minimize cost.
3. **Establish Constraints** – Limitations such as budget, labor, or resources.
4. **Solve the Model** – Use methods like the **Simplex algorithm** or Excel Solver.
5. **Interpret Results** – Analyze outcomes for business decisions.

Example:

A factory optimizing product mix to maximize profit under limited raw materials.

4A. Explain Features of Linear Programming (L3 CO3 – 4M)

- **Objective Function:** Represents the goal (maximize/minimize).
 - **Constraints:** Represent limitations or restrictions.
 - **Linearity:** Both the objective and constraints are linear equations.
 - **Non-negativity:** Decision variables must be zero or positive.
 - **Feasible Region:** All possible solutions that satisfy constraints.
 - **Optimal Solution:** The best solution within the feasible region.
-

5A. Discuss Implementation of Linear Optimization Models on Spreadsheets (L3 CO3 – 4M)

Implementation Steps:

1. **Input Data:** Enter coefficients, variables, and constraints in Excel.
2. **Define the Objective Function:** Create a formula for the target (e.g., total profit).
3. **Use Solver Add-in:**
 - Set objective cell (maximize/minimize).
 - Select decision variable cells.
 - Add constraint equations.
4. **Run Solver:** It finds the optimal solution.
5. **Analyze Results:** Interpret output values and test different scenarios.

Example:

Using Excel Solver to maximize total profit by adjusting production quantities.

6A. Discuss the Solving Procedures Using Linear Optimization Models (L3 CO3 – 4M)

Solving Procedure:

1. **Formulate the Problem:** Define objective function and constraints.
2. **Graphical Method (for 2 variables):** Plot constraints and find the feasible region.
3. **Simplex Method (for multiple variables):** Iterative approach to reach optimal solution.
4. **Computer-Based Methods:** Use tools like Excel Solver, LINDO, or Python libraries.
5. **Sensitivity Analysis:** Test how changes in variables affect the optimal solution.

Outcome:

Provides the **most efficient allocation of resources** to achieve the desired business goal.

Would you like me to prepare a **CO3 + CO4 + CO5 combined revision PDF** (short + essay answers in one printable format)? It'll be perfect for your end-sem exam prep.



Short Answer Questions

Level of
Bloom
Taxonomy

CO Marks
(2M)

1 Difference between Data visualisation and Visual analytics L3 CO3 2M

2 What is linear programming? L2 CO3 2M

3 Explain two optimization rules L2 CO3 2M

4 List two Methods of linear programming L3 CO3 2M

5

Explain:

1. Objective function

2. Non-negative restrictions

L3 CO3
2M

6 List the steps involved in Formulation of problem

Here's your **CO3 – Short Answer Questions (2 Marks each)** summarized in a clean, exam-ready format



1. Difference between Data Visualization and Visual Analytics (L3 CO3 – 2M)

Data Visualization	Visual Analytics
Represents data in visual formats like charts and graphs.	Combines visualization with analytical methods for deeper insights.
Focuses on presenting data clearly.	Focuses on exploring and interpreting data interactively.

2. What is Linear Programming? (L2 CO3 – 2M)

Linear programming (LP) is a **mathematical technique used to optimize** (maximize or minimize) a linear objective function subject to a set of linear constraints.

Example:

Maximize profit given limited raw materials or labor.

3. Explain Two Optimization Rules (L2 CO3 – 2M)

1. **Objective Rule:** Define whether the goal is to **maximize** (e.g., profit) or **minimize** (e.g., cost).
 2. **Constraint Rule:** All **limitations** (resources, budget, time) must be represented as linear equations or inequalities.
-

4. List Two Methods of Linear Programming (L3 CO3 – 2M)

1. **Graphical Method** – Used for two-variable problems; solutions are found visually.
 2. **Simplex Method** – An iterative mathematical approach used for multi-variable problems.
-

5. Explain:

1. Objective Function

It is a **mathematical expression** representing the goal of optimization — either to maximize or minimize (e.g., $\text{Max } Z = 5x + 4y$).

2. Non-Negative Restrictions

Decision variables cannot be negative, meaning $x \geq 0, y \geq 0$, since negative production or cost isn't practical.

6. List the Steps Involved in Formulation of a Problem (L3 CO3 – 2M)

1. Identify the **decision variables**.
 2. Define the **objective function** (maximize or minimize).
 3. Establish the **constraints** (limitations).
 4. Apply **non-negative restrictions**.
 5. Express the problem in **mathematical form** for solving.
-

Would you like me to make a **combined 2-mark CO3 + CO4 + CO5 answer sheet** (in one A4 PDF) for last-minute revision?